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Rust 1987 Model Results

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Estimated Parameters:

Beta (Discount Factor): 0.9999
Theta_11: 2.2930
Theta_3: 0.3919
RC (Replacement Cost): 10.0749

Log-Likelihood:

Log-Likelihood: -163.5843

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Part b)

I am not sure how to integrate the 3D arrays to ease the calculation, so everything is a bit in the air.

For using CCPs, referring to Hotz, Miller, Sanders, Smith (1994), we can use the following logic:

- Estimate CCPs from the data for each state. This is done only once.
- Estimate transition probabilities e.g. AR(1).
- Given theta values recover the continuation values with the estimated CCPs. Then recover lifetime utility.
- Write the likelihood function again and run a maximum likelihood estimation in a similar fashion over theta values.

So we would need to create a module for CCP calculation where:

- $\text{def empirical_CCPs}(x, i) : P(i \text{ given } x) = \text{number of choice } i \text{ given state } y \text{ over all state } y \text{ instances.}$

Then we would need a simulation module for continuation value calculation given the empirical CCPs.

- $\text{def EV_calculator}(x, i) : \text{EV_continuation} = \text{sum over } x [P(y \text{ given } x, i) * V_hat(x)]$
where P can be calculated through x and AR(1 process), V-hat being approximated value function at state x. This estimation can be done through simulation and utilizing transition probabilities and CCPs (i.e. simulate the choices and transition of states for enough of a length and given parameter values and parametric form, iteratively sum the flow utility to get continuation values).

Then we can calculate the total utility with estimated EVs and flow utilities in utility function script. This step can actually be merged with previous one in one function.

Then we can calculate the implied CCPs given model specifications and previous utility calculations

Then we can calculate the log-likelihood function and get its value, similarly in the maximum likelihood module.

Then we can search over the theta, to minimize the -log likelihood value.

The advantages are that this method does not require value function iteration and fixed point existence. However, this method produces simulation error through length of simulation and averaging over instances finitely.